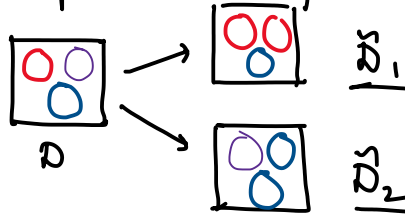


Lecture 22

Saturday, May 2, 2026 9:21 AM

Recap: Bagging (Bootstrap Aggregating)

Bootstrapping: Simulating having multiple datasets by sampling with replacement from the given dataset.



Bagging

1. For $b = 1, 2, \dots, B$
 - a) Create dataset $\tilde{D}_b$ by bootstrapping
 - b) Train model on $\tilde{D}_b$ (say a decision tree)
2. Prediction
 - • Regression: $\hat{y} = \text{average of } B \text{ models}$
 - • Classification: $\hat{y} = \text{majority vote of } B \text{ models}$

Limitations of Bagging with trees as models:
If one feature has very low Gini Impurity index when used as a split, all B trees will use it for the root node.

We want tree to have diversity.

Fix: Random Forests: Bagging + Decision trees + feature subsampling at each node

	Decision trees	Random Forest
Pros	Easy to interpret	Harder to interpret
Cons	poor performance	often much better

Boosting (reference material will be provided)

Supervised learning	Unsupervised learning
- $(x^{(i)}, y^{(i)})$ pairs $x^{(i)} \in \mathbb{R}^d$ $y^{(i)} \in \mathbb{R}$	- $x^{(i)}$ only

- learn $f: x \rightarrow y$

- Discover structure in $\{x^{(i)}\}_{i=1}^N$

Unsupervised learning $\left\{ \begin{array}{l} \text{no ground truth} \\ \text{subjective} \end{array} \right.$

Clustering

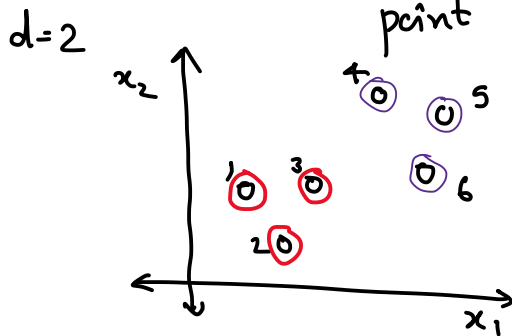
- Defn: Clusters - set of data points with "similar" features. Similarity is measured by a distance metric -

Euclidean distance

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

$$\sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$$

- Clustering: $C: \{1, 2, \dots, N\} \rightarrow \{1, 2, \dots, k\}$
a function that assigns each data point i to a cluster



$k=2$

data	cluster index
1	1
2	1
3	1
4	2
5	2
6	2

Application: $N = 500,000$ customers
 $d = 20 \leftarrow$
 $k = 10$

- Clusters will group similar customers in a group
- Targeted marketing for each group

Formal Problem Setup.

- Dataset $\{x_1, x_2, \dots, x_N\}$ $x_i \in \mathbb{R}^d$
- Integer k : desired number of clusters

Output: Partition of the N points into k non-overlapping groups

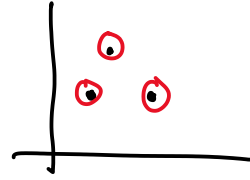
center of each sub-group: average value of the feature vectors of the data-points in that group

Objective = ??

Minimizing within cluster spread

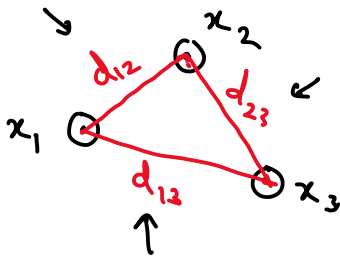
C_k - cluster

m_k - points in the cluster



$$W(C_k) = \frac{1}{m_k} \sum_{i,j \in C_k} \|x_i - x_j\|^2 \quad \Leftarrow$$

$\downarrow$ spread $\downarrow$ # of points in C_k $\underbrace{\hspace{2cm}}$ square of euclidean distance between x_i & x_j



$$W(C_k) = \frac{1}{3} (d_{12}^2 + d_{23}^2 + d_{31}^2)$$

Total Spread : $W(C) = \sum_{k=1}^K W(C_k)$

$\downarrow$ Objective function $\downarrow$ clustering function

To find clusters such that $W(C)$ is minimized

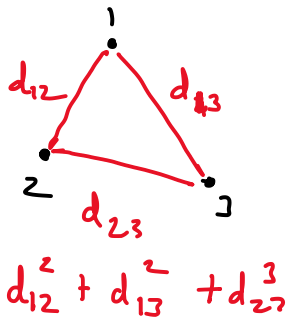
How is k determined? (next class)

Center of data (??)

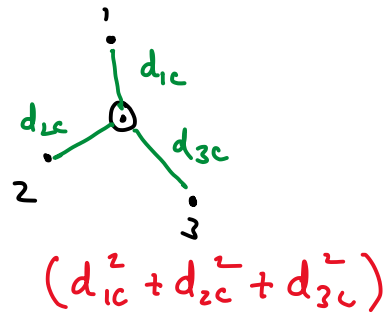
Theorem: • C_k be a ~~class~~ cluster of m_k points.
 • $\mu_k = \frac{1}{m_k} \sum_{i \in C_k} x_i$ ← center centroid

$$\sum_{i,j \in C_k} \|x_i - x_j\|^2 = 2m_k \sum_{i \in C_k} \|x_i - \mu_k\|^2$$

sum of pairwise squared distances



sq. distance of x_i from centroid



Objective.

$$\min W(C) = \min \text{SSE}(C, \mu)$$

$$= \min \sum_{i=1}^N \|x_i - \mu_{c(i)}\|^2$$

point

centroid of the ~~cluster~~ cluster containing $\mu_{c(i)}$

$$\mu_{c_l} = \frac{1}{m_l} \sum_{i \in c_l} x_i$$



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